



Unit 4 · Outcome 2 · Energy Sources

The mine stops. Now someone has to heal the hole.

KK08

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



Why we can't just walk away

A decommissioned energy site isn't neutral ground — it is actively dangerous and degraded. Rehabilitation exists to remove that hazard and hand the land back with a future.

Picture the fictional Yarrabin open-cut, the brown-coal pit that fed the Yarrabin Energy Park for sixty years. When the last shovel-load is burned, it leaves an 18-kilometre-perimeter hole up to 135 metres deep, ringed by near-vertical walls of soft clay and coal. Left alone, that site slumps, erodes, catches fire, and leaches acid and metals into groundwater. Nothing lives there because there is no soil — mining stripped it decades ago.

Australian law forces the operator to fix this. The regulator's benchmark is a single, examinable phrase: the land must be returned to a safe, stable and sustainable landform. Safe = no fire, collapse or toxic exposure risk to people. Stable = the landform won't keep moving, slumping or eroding. Sustainable = it can support a self-maintaining land use (bush, farmland, a lake, recreation) without endless human propping-up.

◆ ANALOGY

The deep-wound analogy Healing a mine is like healing a deep cut. First a surgeon physically closes and stabilises the wound — that's the mechanical work (stitches, setting bone). Only then can the body's living tissue regrow across it — that's the biological work (new skin, blood supply). Skip the stabilising and the living layer just sloughs straight off. Same order, every time.

MECHANICAL · PHYSICAL & ENGINEERING

Two toolkits: mechanical & biological

VCAA splits rehabilitation into exactly two process families. Getting a clean definition of each — and never blurring them — is the core skill this dot point tests.

Non-living, engineering processes that reshape and stabilise the landform and lock away hazards. Bulldozers, excavators, dump trucks and graders do the work.

Processes that use living organisms — plants, microbes and fungi — to rebuild soil, vegetation and a working ecosystem on top of the stabilised landform.

• NOTE

The one-line test: if it's done by a machine to the shape of the land → mechanical. If it's done by something alive, or relies on something alive → biological. Topsoil is the bridge: moving it is mechanical, but what's in it (the living seed bank and microbes) is biological.

LIVE 3D DRAG TO ORBIT · CLICK A LAYER 3D LAYER CROSS-SECTION (WEBGL UNAVAILABLE — USE THE LAYER BUTTONS BELOW). FULL PROFILE — 5 ENGINEERED LAYERS

Inside a rehabilitated landform

◇ ACTIVE RECALL

A rehabilitated site is built up in engineered layers. Peel them back to see where the mechanical work ends and the biological work begins.

- **NOTE**

The profile, bottom to top: waste-rock backfill rebuilds the bulk landform (mechanical) → a low-permeability clay seal caps the coal so it can't burn or leach (mechanical) → subsoil is deep-ripped and shaped (mechanical) → stored topsoil is respread, reawakening its seed bank and microbes (the bridge) → vegetation roots in and the living ecosystem returns (biological).

Why the order is non-negotiable

A favourite exam discriminator: explaining the sequence. Mechanical work almost always comes first — and there's a clear cause-and-effect reason.

You cannot grow a forest on a 70° wall of slumping clay. The mechanical stage must come first to create a landform that won't move: slopes are battered back to a stable angle, the void is backfilled, drainage is shaped, and the hazardous coal is sealed under clay. Only once the ground is safe and stable does the biological stage make sense — because revegetation needs a surface that stays put long enough for roots to establish.

The crucial hinge between the two is topsoil. Good operators strip and stockpile topsoil before mining begins, then respread it at the end. That single move is the most powerful step in the whole process, because living topsoil already contains the seed bank, microbes and fungi an ecosystem needs — you're not building life from scratch, you're reawakening it.

- ◆ **ANALOGY**

Foundations before furniture No one furnishes a house before the slab is poured and the frame is up. Mechanical = the structure (foundation, frame, weatherproofing). Biological = moving in (the living, functioning home). Do them out of order and everything you place inside collapses with the building.

- **NOTE**

Progressive rehabilitation is the gold standard: rehabilitating worked-out sections while the rest of the site is still being mined, rather than waiting until the very end. It spreads cost and risk and means freshly-stripped topsoil goes straight back down while it's still biologically alive.

LIVE 3D DRAG TO ORBIT 3D REVEGETATION / SUCCESSION SCENE (WEBGL UNAVAILABLE — THE YEAR SLIDER STILL DRIVES THE READOUT). YEAR 0 · BARE RESPREAD TOPSOIL

Bringing the land back to life

- ◆ **ACTIVE RECALL**

The biological stage is slow, living and measurable. Watch a stabilised surface move through ecological succession — and see how plants can actually clean the ground.

- **NOTE**

Succession. Pioneer grasses and nitrogen-fixers colonise first, building soil organic matter; shrubs follow; then canopy trees. Returning soil microbes and mycorrhizal fungi are the hidden engine — scientists actually measure the microbial community to judge whether rehabilitation is succeeding.

◇ ACTIVE RECALL

Direct-seeding and planting native, locally-provenanced species so the cover is self-sustaining and matches surrounding bush. Microbes break down hydrocarbons and other contaminants in soil and water — e.g. microalgae stripping dissolved metals from mine water. Specific plants (like vetiver grass) draw up or stabilise heavy metals through their roots, pulling contamination out of the ground.

Rehabilitate is not restore

This distinction is quietly worth marks. Examiners punish students who treat the two as synonyms.

Returning the land to a safe, stable and sustainable condition fit for a viable land use — which may be different from the original (e.g. a former pit becomes farmland, a recreation lake, or a solar farm). The bar is function and safety, not historical accuracy.

The more ambitious goal of returning the site to its original, pre-disturbance ecosystem — same species, same structure, same function. Far harder, often impossible after deep open-cut mining where the original soil profile is gone forever.

• NOTE

Why it matters: after 135 m of open-cut mining the original landform and soil simply no longer exist, so full restoration is usually unrealistic. Most real energy-site projects target rehabilitation to a chosen end land use — which is why the pit-lake option is even on the table.

The Latrobe Valley — Australia's real test case

Yarrabin is fictional; the science and the stakes are not. Victoria is right now attempting the largest mine rehabilitation in its history, on our doorstep in Gippsland.

Three giant brown-coal mines — Hazelwood (closed 2017), Yallourn (closing 2028) and Loy Yang (around 2035) — have powered Victoria since the 1920s. Hazelwood alone is an 18 km-perimeter void up to 135 m deep, with rehabilitation estimated at roughly \$743 million. The Hazelwood mine is bigger than the Melbourne CBD; cleaning it up is the biggest rehabilitation Victoria has ever attempted.

The leading plan for each is a pit lake — slowly filling the void with water to keep the walls stable and the coal submerged so it can't burn. Yallourn's void alone needs an estimated ~630 gigalitres, drawn over decades from the stressed Latrobe/Morwell river system — which is exactly why it's controversial. The mechanical work (battering walls, sealing coal, managing water) and biological work (revegetating the rim and catchment) run together over generations.

The Mineral Resources (Sustainable Development) Act 1990 requires every licensee to rehabilitate to a safe, stable, sustainable landform. The 2026 Financial Assurance Amendment (passed Feb 2026) adds a trailing liabilities scheme so a company can't dodge the cleanup bill.

The Mine Land Rehabilitation Authority oversees Victoria's three declared mines, and Resources Victoria runs the Latrobe Valley Regional Rehabilitation Strategy. Plans go through a full Environment Effects Statement.

Rehabilitation bonds make the operator pre-pay security for the cleanup — a direct application of the user-pays principle. Hazelwood's bond (~\$289 m) sits well below the estimated full cost, a live point of public debate.

How it ties to the sustainability principles

KK08 doesn't live in isolation — examiners love linking rehabilitation back to the principles you met earlier in the course.

Rehabilitating now means we don't dump a dangerous, useless landscape on future generations. An un-rehabilitated pit is a debt passed forward.

Revegetation, bioremediation and reconnecting habitat rebuild functioning ecosystems and biodiversity on degraded land.

Rehabilitation bonds and trailing-liability laws force the company that profited from the energy to fund the cleanup, not taxpayers.

Stabilising walls and sealing coal before failure — and assessing pit-lake water impacts up-front via an EES — acts ahead of proven harm.

Command-word decoder

◆ COMMAND WORDS

The verb in the question tells you what shape your answer must take. Tap each to see what the examiner actually wants for this topic.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

Six exam traps

⚠ EXAM TRAPS

The predictable ways students drop marks on this dot point. Open each before you sit a SAC.

P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

P oint · E vidence/example · L ink. Write a paragraph and watch live which key ideas you've actually hit. Aim to light up every chip.

Worked answer: weak vs strong

Same question, two responses. Read why the examiner's pen moves — then steal the strong version's structure.

• NOTE

Question (4 marks): Describe one mechanical and one biological process used to rehabilitate a former open-cut coal mine, and explain how each contributes to a safe, stable and sustainable landform.

They fix up the mine by cleaning it and planting some trees so it looks nice again and animals can come back to live there like before.

A mechanical process is battering down (regrading) the steep highwalls and backfilling the void with overburden, which creates a stable landform that won't slump or collapse — making it safe and stable. A biological process is resspreading stockpiled topsoil and revegetating with native species, whose roots bind the surface against erosion and rebuild a self-sustaining ecosystem — making the landform sustainable.

Practice exam · 22 marks

★ PRACTICE & SELF-MARK

Five VCAA-style questions. Attempt each in your logbook first, then open the model answer and self-mark honestly with the stars. Your score builds in the dock.

One-screen cheat sheet

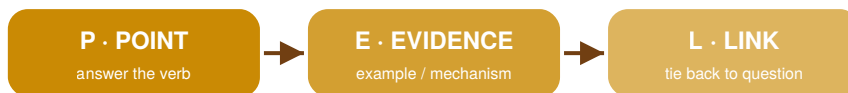
- Reshape / batter steep walls to stable slope
- Cap / seal coal with clay (stop fire & leaching)
- Deep-rip to relieve compaction
- Contour drainage; build erosion banks
- Plug / cap oil & gas wells
- Respread topsoil (seed bank + microbes)
- Revegetate with native species
- N-fixers & mycorrhizal fungi rebuild soil
- Bioremediation — microbes clean contaminants
- Phytoremediation — plants draw up metals
- Reintroduce fauna; manage weeds
- Goal = safe, stable, sustainable landform
- Mechanical first, then biological
- Topsoil = the bridge between them

- Rehabilitate ≠ restore (original ecosystem)
- Progressive rehab = clean as you mine
- Links: intergen. equity, user-pays (bonds)

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

They fix up the mine by cleaning it and planting some trees so it looks nice again and animals can come back to live there like before. "Cleaning it" isn't a named mechanical process — no reshaping, backfill or capping. "Planting trees" is on the right track but vague, and never linked to the goal. "Like before" wrongly implies full restoration; ignores safe/stable/sustainable.

✓ STRONG / MODEL ANSWER

A mechanical process is battering down (regrading) the steep highwalls and backfilling the void with overburden, which creates a stable landform that won't slump or collapse — making it safe and stable. A biological process is respreading stockpiled topsoil and revegetating with native species, whose roots bind the surface against erosion and rebuild a self-sustaining ecosystem — making the landform sustainable. Names a specific mechanical process AND its function (stability/safety). Names a specific biological process AND its function (sustainability). Explicitly ties both to the marked phrase "safe, stable and sustainable".